

Damped Harmonic Motion

Lecture: 9

Frictional Force:

$$\vec{F}' \propto -v$$

$$F' = b v$$

For horizontal force:

$$m a = -k x - b v$$

$$m \frac{d^2 x}{dt^2} = -k x - b \frac{dx}{dt}$$

$$\frac{d^2 x}{dt^2} = -\frac{k}{m} x - \frac{b}{m} \frac{dx}{dt}$$

$$\frac{d^2 x}{dt^2} = -\omega^2 x - \gamma \frac{dx}{dt}$$

$$\omega^2 = \frac{k}{m}$$

damping constant

$$\gamma = \frac{b}{m}$$

$$\frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + \omega^2 x = 0$$

→ equation of Damped Harmonic motion. (D.H.M)

→ For this part it is a D.H.M.

Trial Solution:

$$\frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

Let, $x = B e^{pt}$ $B = \text{Amplitude}$

$$\frac{dx}{dt} = B p e^{pt}$$

$$\frac{d^2x}{dt^2} = B p^2 e^{pt}$$

$$B p^2 e^{pt} + \gamma B p e^{pt} + \omega_0^2 B e^{pt}$$

$$p^2 + \gamma p + \omega_0^2 = 0 \quad \text{--- ①}$$

Comparing with $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\text{So, } p = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2}$$

$$= -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2 - 4\omega_0^2}{4}}$$

$$p = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

Condition 1: $\omega_0^2 > \frac{\gamma^2}{4}$ (oscillatory)

Condition 2: $\omega_0^2 = \frac{\gamma^2}{4}$ (critical)

Condition 3: $\omega_0^2 < \frac{\gamma^2}{4}$ (over)

$$\omega_0^2 > \frac{\gamma^2}{4} \text{ (oscillatory)}$$

$$x = A e^{-\frac{\gamma}{2}t} \cos(\omega_1 t + \phi)$$

Oscillatory solution, $A \cos(\omega_1 t + \phi)$ multiplied by a damping factor $e^{-\frac{\gamma}{2}t}$. As $\gamma \rightarrow 0$ we approach our undamped oscillator.

$$x = 5 e^{-3t} \cos(7t)$$

Here, $x = y$, $A = 5$, $-\frac{\gamma}{2} = -3$, $t = \tau$, $\omega_1 = 7$.

$$\omega_0^2 = \frac{\gamma^2}{4} \text{ (Critical damping)}$$

$$x = (A + Bt) e^{-\frac{\gamma}{2}t}$$

γ বাড়লে oscillation (দোলন) কমে যেতে থাকবে। যত দ্রুত $\omega_0^2 = \frac{\gamma^2}{4}$ আর oscillation ঘটে না। (Fast approach)

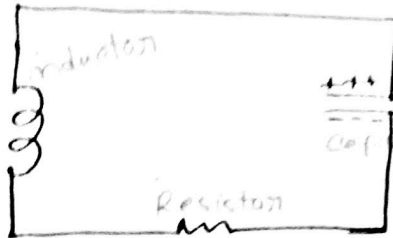
$$\omega_0^2 < \frac{\gamma^2}{4} \text{ (over damping)}$$

$$x = B_1 e^{(-\frac{\gamma}{2} + \alpha)t} + B_2 e^{(-\frac{\gamma}{2} - \alpha)t}$$

A slower approach to the rest position is observed.

RLC Circuit

$$V_L = L \frac{di}{dt}$$



$$V_C = \frac{Q}{C}$$

* Some energy will be absorbed by resistor.
* discharged to charged.

According to KVL:

$$V = V_L + V_C$$

$$\Rightarrow iR + L \frac{di}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{R}{L} \frac{dQ}{dt} + \frac{1}{LC} Q = 0$$

$$\boxed{\frac{d^2Q}{dt^2} + \gamma \frac{dQ}{dt} + \omega_0^2 Q = 0}$$

$$\boxed{\gamma = \frac{R}{L}}$$

$$\boxed{\omega_0^2 = \frac{1}{LC}}$$

Compare to $\omega_0^2 > \frac{\gamma^2}{4}$:

$$\frac{1}{LC} > \frac{R^2}{4L^2} \text{ (Oscillatory)}$$

$$\frac{1}{LC} = \frac{R^2}{4L^2} \text{ (Critical)}$$

$$\frac{1}{LC} < \frac{R^2}{4L^2} \text{ (Over)}$$

$$\frac{1}{LC} > \frac{R^2}{4L^2}$$

$$Q = A e^{-\frac{R}{2L}t} \cos(\omega_1 t + \phi)$$

$$\omega_1 = \sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

☐ A capacitor $1\mu\text{F}$, an inductor 0.2h and a resistance 800Ω are joined in series. Is the circuit oscillatory? Find the frequency of oscillation.

Solve:

$$\frac{1}{LC} = \frac{1}{1 \times 10^{-6} \times 0.2} = 5 \times 10^6$$

$$\frac{R^2}{4L^2} = \frac{(800)^2}{4(0.2)^2} = 4 \times 10^6$$

$$\left. \begin{array}{l} C = 1\mu\text{F} \\ = 1 \times 10^{-6}\text{F} \\ L = 0.2\text{h} \\ R = 800\Omega \end{array} \right\}$$

$\frac{1}{LC} > \frac{R^2}{4L^2}$ So, The circuit is oscillatory.

$$f = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \frac{1}{2\pi} \sqrt{5 \times 10^6 - 4 \times 10^6}$$

$$= 159.15\text{Hz}$$